Aria Koul

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Summary

Master of Information and Data Science student with experience in machine learning, data analysis, and Python programming. Demonstrated ability to apply data science techniques to real-world problems. Proficient in data visualization and multiple programming languages. Seeking data science positions.

Education

University of California, Berkeley
Jan 2026 – Expected Aug 2027

Information and Data Science | M.S.

Relevant coursework: Applied Machine Learning · Machine Learning Systems Engineering · Data Engineering · Data Visualization

University of California, Santa Cruz
Oct 2020 – Jun 2024

Physics (Astrophysics) | B.S.

GPA: 3.72

Minors: Statistics

Relevant coursework: Physics and Machine Learning · Astrophysics Advanced Laboratory · Classical and Bayesian Inference

Work Experience

Research Assistant
Jul 2023 – Nov 2025

UC Santa Cruz

- Implemented a program to model cosmic rays' trajectories in the magnetosphere.
- Converted various physics algorithms and equations into Python.
- Analyzed 5000+ entry datasets using pandas, numpy, matplotlib, and scipy.

Engineering Intern
Apr 2025 – Jul 2025

Picarro

- Analyzed stability of new spectroscopy instrument using Python for 90-day prototype data using various statistical measures and methods.
- Debugged and tested spectroscopy analyzer data.

AI Engineering Intern
Dec 2024 – Mar 2025

Airlinq

- Created an AI chatbot agent using Python and the OpenAI API platform for customers to chat with regarding launching connected car services.
- Used Flask, LangChain, pandas, datetime, dotenv, and logging.

Projects

Predicting Next Day Wildfire Spread in the United States
Jul 2026

- Created three progressively deeper convolutional neural networks using TensorFlow that predict next day wildfire spread using satellite imagery of environmental conditions.
- Performed feature engineering to aid with exploratory data analysis.
- Optimized models to increase AUC-ROC by 30%.

Classifying Cosmic Rays

Mar 2024

- Developed a machine learning algorithm using PyTorch neural networks to predict the type of particle in a primary cosmic ray, achieving a 90% accuracy rate.
- Optimized hyperparameters to reduce training time by 50%.

Stock Market Analysis

Mar 2022

- Implemented a stock performance evaluation tool using Python leveraging overnight and day gains of any stock over any time period, resulting in optimized investment strategies.
- Created a suite of unit tests to ensure reliability.

Skills

Python, Machine Learning, Data Analysis, Data Visualization, SQL, HTML, CSS, MATLAB, Java

Awards

Phi Beta Kappa Academic Honor Society

Member

Publications

Through the heliospheric lens: Directional deflection of high-energy cosmic-ray electrons and positrons. *Astroparticle Physics* — Jan 2026